



TECHNICAL DATA SHEET

— Parasmani HVAC/R Copper Tubes

Seamless Copper Tubes for HVAC, Refrigeration and VRF / VRV Systems

I. SCOPE & APPLICABLE STANDARDS

This data sheet covers **Parasmani** Cu-DHP seamless round copper tubes manufactured for air-conditioning and refrigeration service, including VRF / VRV systems, split air-conditioning, general HVAC, refrigeration and industrial process piping. Tubes are supplied as straight lengths in hard drawn, light drawn (half hard) and light annealed (quarter hard) condition, and as soft annealed pancake coils.

Standard	Application
IS 10773 : 2025	Wrought copper tubes for refrigeration and air-conditioning purposes (Second Revision) — primary Indian standard for the tube
JIS H 3300	Copper and copper alloy seamless pipes and tubes — C1220T
ASTM B 280	Seamless copper tube for air-conditioning and refrigeration field service
ASTM B 68	Seamless copper tube, bright annealed
ASTM B 75	Seamless copper tube
ASTM B 88	Seamless copper water tube

Where a project specification calls for a standard other than the above, requirements shall be confirmed in writing prior to manufacture.

2. MATERIAL & CHEMICAL COMPOSITION

All copper tubes are manufactured ensuring superior purity and consistent metallurgical properties. Tubes for HVAC/R service are produced from C12200 DHP grade copper — designated Cu-DHP under IS 10773 : 2025.

Specification	Copper Alloy No. / Grade	Cu %	P %
IS 10773 : 2025	Cu-DHP	99.90% Min. (Cu + Ag)	0.015 – 0.040%
JIS H 3300	C12200	99.90% Min.	0.015 – 0.040%
ASTM B 68	C12200	99.90% Min.	0.015 – 0.040%
ASTM B 280	C12200	99.90% Min.	0.015 – 0.040%
ASTM B 75	C12200	99.90% Min.	0.015 – 0.040%



The internal bore is compatible with prominent modern refrigerants including R-410A, R-32, R-22, R-134a, R-600 and R-600a.

3. MECHANICAL PROPERTIES

Standard	Condition / Temper	Tensile (MPa)	Elong. %	Grain Size (mm)	Hardness	Bore Cleanliness
IS 10773:2025	Soft annealed	210 Min.	40	0.040 Min.	—	0.038 g/m ² Max.
IS 10773:2025	Light annealed	220 Min.	40	0.015 – 0.040	—	0.038 g/m ² Max.
IS 10773:2025	Light drawn	245 Min.	20	N/A	—	0.038 g/m ² Max.
IS 10773:2025	Hard drawn	290 Min.	—	N/A	—	0.038 g/m ² Max.
JIS H 3300	Soft annealed (O)	205 Min.	40	0.025 – 0.060	60 HV Max.	—
JIS H 3300	Half hard	245 – 325	—	—	70 – 110 HV	—
JIS H 3300	Hard	315 Min.	—	—	100 HV Min.	—
ASTM B 68	Soft annealed (O60)	210 Min.	40	0.040 Min.	—	—
ASTM B 280	Soft annealed (O60)	205 Min.	40	0.040 Min.	—	0.038 g/m ² Max.
ASTM B 280	Hard (H58)	250 Min.	—	—	—	—
ASTM B 75	Half hard (H55)	250 – 325	—	—	30 – 60 HR 30T	—
ASTM B 75	Hard (H80)	310 Min.	—	—	55 Min. HR 30T	—

4. PRODUCT RANGE

Straight Copper Tubes

Condition / Temper	Hard drawn, light drawn (half hard), light annealed (quarter hard)
Size Range (Outside Diameter)	6 mm to 130 mm
Wall Thickness	0.70 mm to 2.0 mm
Length	Up to 6000 mm



Pancake Copper Coils

Condition / Temper	Soft annealed
Size Range (Outside Diameter)	4.76 mm to 22.23 mm
Wall Thickness	0.4 mm to 1.2 mm
Length	3000 mm to 16000 mm

Hard drawn tubes are available in sizes up to 5½" OD, wall thicknesses up to 10 SWG and lengths up to 6 metres. Soft annealed tubes are available in sizes up to 7/8" OD, wall thicknesses up to 18 SWG and lengths from 3 to 16 metres. Tubes are also manufactured to specific customer requirements and project specifications.

5. DIMENSIONAL TOLERANCES

Dimensional tolerances are as specified in IS 10773 : 2025. Outside diameter, wall thickness and length are otherwise as mutually agreed between purchaser and supplier.

5.1 Tolerance on Mean Wall Thickness

TOLERANCES ON MEAN WALL THICKNESS (mm) BY OUTSIDE DIAMETER BAND					
Wall Thickness, WT (mm)	3.0 < OD ≤ 16	16 < OD ≤ 25	25 < OD ≤ 50	50 < OD ≤ 100	100 < OD ≤ 180
0.25 < WT ≤ 0.40	± 0.03	± 0.04	± 0.05	—	—
0.40 < WT ≤ 0.60	± 0.05	± 0.05	± 0.06	—	—
0.60 < WT ≤ 0.90	± 0.06	± 0.06	± 0.08	± 0.10	—
0.90 < WT ≤ 1.5	± 0.08	± 0.09	± 0.09	± 0.10	± 0.20
1.5 < WT ≤ 2.0	± 0.09	± 0.10	± 0.10	± 0.15	± 0.20
2.0 < WT ≤ 3.0	± 0.10	± 0.10	± 0.10	± 0.20	± 0.20

5.2 Tolerance on Mean Outside Diameter

Outside Diameter, OD (mm)	Tolerance (mm)
3.0 < OD ≤ 16	± 0.05
16 < OD ≤ 25	± 0.06
25 < OD ≤ 50	± 0.08
50 < OD ≤ 75	± 0.10
75 < OD ≤ 100	± 0.12
100 < OD ≤ 125	± 0.15
125 < OD ≤ 150	± 0.18



5.3 Tolerance on Length of Tubes Supplied in Straight Length

Length, l (mm)	Tolerance (mm)
$l \leq 600$	+ 1.6 / - 0
$600 < l \leq 2\,000$	+ 2.0 / - 0
$2\,000 < l \leq 4\,000$	+ 5.0 / - 0
$4\,000 < l \leq 7\,000$	+ 8.0 / - 0
$l > 7\,000$	+ 10.0 / - 0

Minus tolerances on length are not permissible.

5.4 Roundness, Ovality and Eccentricity

Ratio t / D	Roundness Tolerance
Up to 0.03	1.5%
Over 0.03 to 0.05	1.1%
Over 0.05 to 0.10	0.8% or 0.5 mm, whichever is more
Over 0.10	0.7% or 0.05 mm, whichever is more

t = Nominal wall thickness; D = Nominal outside diameter. These tolerances apply only to light and hard drawn tubes.

6. DIMENSIONAL & PRESSURE DATA

6.1 VRV / VRF Copper Tubes — JIS H 3300

HARD DRAWN STRAIGHT COPPER TUBES STANDARD PIPE LENGTH: 3 MTR								
ø (in)	ø (mm)	Temper	W.T. (mm)	≈ SWG	kg / Mtr.	kg / Pipe	Max Working (MPa)	Bursting (MPa)
1/4"	6.35	Hard	0.80	21	0.125	0.380	44.14	88.27
3/8"	9.53		0.80	21	0.197	0.600	28.35	56.69
1/2"	12.70		0.80	21	0.268	0.810	20.90	41.79
5/8"	15.88		0.80	21	0.339	1.020	16.54	33.07
5/8"	15.88		1.00	19	0.418	1.260	20.89	41.78
3/4"	19.05		0.80	21	0.411	1.240	13.69	27.38
3/4"	19.05		1.00	19	0.507	1.530	17.26	34.52
7/8"	22.23		0.80	21	0.482	1.450	11.67	23.34
7/8"	22.23		1.00	19	0.597	1.800	14.70	29.40
1"	25.40		0.88	20	0.607	1.830	11.23	22.45
1"	25.40		1.00	19	0.686	2.060	12.81	25.61
1 1/8"	28.58		1.00	19	0.775	2.330	11.34	22.68
1 1/4"	31.75		1.10	18.5	0.947	2.850	11.23	22.45
1 3/8"	34.93		1.21	18	1.146	3.440	11.23	22.45
1 1/2"	38.10		1.32	17.5	1.364	4.100	11.23	22.45
1 5/8"	41.28		1.43	17	1.601	4.810	11.23	22.45



6.2 VRV / VRF Copper Tubes — JIS H 3300

SOFT ANNEALED PANCAKE COILS STANDARD COIL LENGTH: 15.24 MTR								
ø (in)	ø (mm)	Temper	W.T. (mm)	≈ SWG	kg / Mtr.	kg / Coil	Max Working (MPa)	Bursting (MPa)
1/4"	6.35	Soft (O)	0.80	21	0.125	1.900	19.15	57.44
3/8"	9.53		0.80	21	0.197	3.000	12.30	36.90
1/2"	12.70		0.80	21	0.268	4.100	9.07	27.20
5/8"	15.88		0.80	21	0.339	5.150	7.17	21.52
5/8"	15.88		1.00	19	0.418	6.350	9.06	27.19
3/4"	19.05		1.00	19	0.507	7.750	7.49	22.47

6.3 Non VRV / VRF Copper Tubes — ASTM B 68

SOFT ANNEALED PANCAKE COILS STANDARD COIL LENGTH: 15.24 MTR								
ø (in)	ø (mm)	Temper	W.T. (mm)	≈ SWG	kg / Mtr.	kg / Coil	Max Working (MPa)	Bursting (MPa)
1/4"	6.35	Soft (O)	0.61	23	0.099	1.500	14.22	42.66
3/8"	9.53		0.66	23	0.165	2.500	10.02	30.06
1/2"	12.70		0.68	22	0.230	3.500	7.65	22.94
5/8"	15.88		0.69	22	0.295	4.500	6.15	18.46
3/4"	19.05		0.71	22	0.366	5.600	5.25	15.75
3/4"	19.05		0.80	21	0.411	6.250	5.94	17.82

W.T. = wall thickness · SWG = Standard Wire Gauge (approximate equivalent) · ø = outside diameter.

Bursting pressure is calculated by the modified Barlow equation, $P = 2 \times UTS \times t / (D - 0.8 \times t)$, where P is bursting pressure in MPa, UTS is ultimate tensile strength in MPa, t is wall thickness in mm and D is outside diameter in mm. Safety factor: hard temper 2, soft annealed 3. Maximum working pressure = bursting pressure ÷ safety factor. Conversion: 1 MPa = 145.038 psi.

7. MANUFACTURING & INTERNAL CLEANLINESS

Tubes are drawn in-house from tube shell, billet or bar. No tube is manufactured from used tube material. The process sequence is:

Copper melting and casting → extrusion → drawing → cutting → straightening / coiling → annealing → visual inspection → eddy current and other tests → packing and dispatch

Production complies with the applicable standards listed in Section 1. Every stage, from raw material inspection to final dispatch, follows strict quality control, including dimensional, mechanical, surface and internal cleanliness checks. Tubes are cut to size, duly drafted, and sealed at both ends.

Tubes are internally cleaned and tested to ensure residue after solvent evaporation does not exceed 0.038 g/m² of interior surface, ensuring optimal refrigeration performance.



8. NINE-STEP QUALITY ASSURANCE

Each lot is subjected to the following in-house test regime before despatch:

01	Chemical composition analysis	06	Drift expanding test
02	Hydrostatic pressure test	07	Flattening test
03	Eddy current test	08	Hydrogen embrittlement test
04	Freedom from defects	09	Internal cleanliness test
05	Pneumatic pressure test		

9. MARKING & PACKING

Every tube carries a permanent surface marking together with two Parasmani authentication holograms. The marking sequence is:

PARASMANI · IS 10773 : 2025 · Cu-DHP · HARD DRAWN · 12.70 mm × 0.80 mm · PT120875

Read in order: brand name · standard · grade · condition · dimensional details · lot number. The lot number is traceable to the mill test certificate issued with the consignment.

10. CERTIFICATION & DOCUMENTATION

- A Manufacturer's Test Certificate accompanies every despatch, giving the lot number, grade, condition, size, thickness and the corresponding chemical composition and physical properties, traceable to the lot number marked on the tube.
- Quality systems operate under ISO 9001:2015 certified processes; scope: manufacture of copper tubes and fittings for use in air conditioning and refrigeration systems.
- Tubes are manufactured under BIS product certification licence and may carry the Standard Mark.

II. PRODUCT ASSURANCE

All copper tubes supplied against order shall conform to the standards, chemical composition, mechanical properties, dimensional tolerances and cleanliness requirements stated in this data sheet. Any deviation required by a project specification shall be mutually agreed in writing prior to manufacture. This sheet is issued for technical reference; commercial terms, pricing and delivery are covered separately in our quotation.

A project-specific technical submittal on our letterhead, together with mill test certificates, is issued on request.